

Every edit stops the playtest

THE USER

Technical designers at small studios iterate on playable prototypes with real playtesters in the room.

THE COST

Every stop costs the thing they are trying to observe: how people actually play.

Beetle's promise

Change the game without stopping the game.

TODAY'S LOOP



Each turn of the loop loses the live playtest you were trying to watch.

Describe any game: Beetle builds it in 2D or 3D, then keeps changing it while people play



1

Describe any game

"hold the frozen hill ten seconds" or "a tower defense where bees protect a hive"

2

2D or 3D, and named

mapped to a 2D genre or a 3D mode and biome; Beetle says which it chose

3

Playable at once

3D: the world starts where the players are. 2D: sprites, tiles, sound and music made for it

4

Checked before it lands

game code validates every spec and patch; the model only proposes JSON

5

Keep changing it while they play

edits in plain words; in 3D they land while players move, no reset

3D WORLDS (THE EXISTING PIPELINE)

- 5 engine modes: relic_hunt, time_trial, king_of_the_hill, checkpoint_race, survival. The model picks one and fills its parameters; it never writes rules.
- 5 biomes, 2 terrains, water or lava, a rising hazard, 3 to 7 m/s, 11 decoration types.
- Streams ahead of players near an edge, up to 24 islands and 48 bridges; the director can switch it off.
- Live validated edits (set_mode, set_biome, set_movement, structural ops), each a new version. Phones are the controllers.

BEETLE 2D STUDIO (/2d)

- Platformer, top-down, puzzle, defense, runner, arena and other level-based games, generated by the same local model through the server's loopback endpoint.
- Procedural pixel sprites, tiles, sound and music, made for the game; no asset pack needed.
- Edits in plain words: known requests are patched directly, the rest are proposed by the model as patch ops.
- The Game Spec is validated by game code and repaired or refused with the exact list. Export as a single-file HTML game.

Adapted from one-prompt game generators: the one-prompt front door and instant playable output. Added: 2D and 3D from one prompt on one local model, live editing without a reset, deterministic validation and repair before publish, local-only inference, phones as controllers (docs/COMPETITIVE.md). A request outside a library lands on the nearest genre or mode and Beetle says so.

The working video



[VIDEO PLACEHOLDER]

Insert the rehearsal recording here (90 to 120 s). Not yet recorded at 13:40 CDT.

The caption names the agent mode used; any time compression is labelled on screen

ON-SCREEN CAPTIONS

0 to 6 s Beetle. Change the game without stopping the game.

6 to 22 s A local model composes the world. Validated by game code before it is published.

22 to 34 s Two players join from their phones. Same screen, their phones only send controls.

34 to 58 s Edits land live. No disconnect, no reset.

58 to 90 s The validator refuses a world nobody can finish (DISCONNECTED_GOAL). The agent repairs it and commits only a passing version.

90 to 105 s Measured on one GB10. Local model, local agent in [direct or OpenClaw] mode, no cloud.

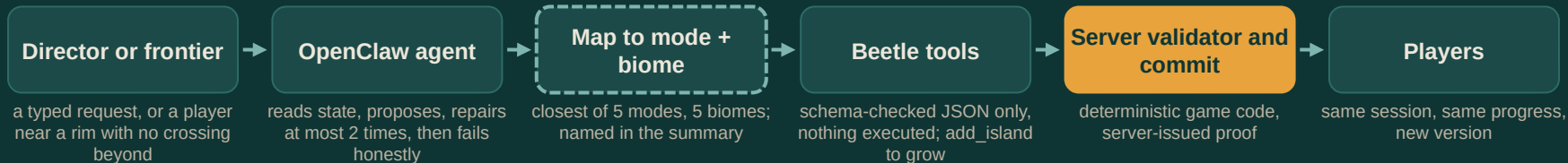
105 to 115 s Beetle

Fresh world, two phones, a live edit, preserved progress, a conflicting edit caught by the validator, the repaired version committed.

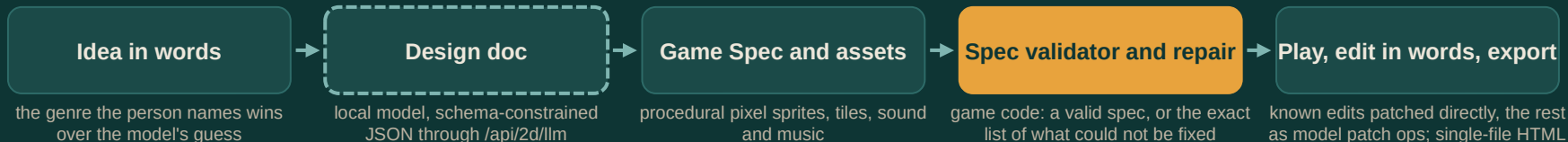
Two engines, one local model, one validation discipline



3D WORLDS: REQUEST TO COMMITTED VERSION



BEETLE 2D STUDIO (/2d): IDEA TO PLAYABLE GAME



ONE LOCAL MODEL

- Both engines call the same configured model (qwen3.5:4b) on Ollama at 127.0.0.1; no cloud fallback host exists.
- 2D model calls go through the server's loopback endpoint, director token required.

ONE VALIDATION DISCIPLINE

- The model only proposes JSON; game code decides. An LLM saying valid is never approval.
- 3D: geometry, sockets, reachability, MODE_INVALID, commit proof. 2D: strict Game Spec checks, repair or refuse.

3D STATE AND GROWTH

- WorldSpec is structure; SessionState is players, relics, score. The agent never writes SessionState.
- Extensions add 1 to 2 islands, at least 12 s apart, up to 24 islands and 48 bridges.

Measured local-first results



MODEL

qwen3.5:4b, Q4_K_M

4.7B, 3.4 GB, Ollama 0.35.1 on loopback.
qwen3.8:27b not benchmarked: pull failed
once, second pull stopped.
docs/MODEL_SELECTION.md

BRIEF, WARM, QUIET GPU

26.2 / 26.5 / 27.7 s

min / p50 / max, 3 runs, first draft valid
(bench-...1791049348895.json). 10 s target
not met. 14 s once after normalization
(RESULTS.md run 3); 33.4 s via OpenClaw.

MODE BRIEFS

18 of 18 right mode

committed briefs 13.0 to 44.1 s; exact mode
missed twice; 8 of 14 final-prompt attempts
committed, failures all geometry; mode and
biome edits 3.0 s (RESULTS.md 'Game
modes from one prompt', direct).

COLD / CONTENTED GPU

23.1 s load; 58 to 214 s

cold: 23.1 s load in a 54.4 s first draft (probe-
run.log). Contended drafts 58.8 to 214.1 s,
one edit 45.1 s
(bench-...1791050596040.json).

EDIT TO COMMIT

2.0 / 8.0 / 14.0 s

min / p50 / max, 6 edits, 5 of 6 committed;
run 3 after normalization, quiet GPU, direct
(run-1791052136845.json). Run 2 before: 2
of 6.

EDIT WHILE 2 PLAY

8.0 s, 15 of 15

acceptance run 2: both controllers walking,
lava + bridge committed, relic and score
unchanged, sockets open, max tick gap 67
ms (ACCEPTANCE.md).

INVALID EDITS CAUGHT

6 of 12 refused

runs 2 and 3 (RESULTS.md); repairs that
then committed: 1; invalid worlds committed:
0. garden5: DISCONNECTED_GOAL
refused 1.7 s, repaired and committed 3.0 s.

NORMALIZATION EFFECT

2 of 30 to 13 of 30

corpus validity without vs with the
deterministic normalizer
(MODEL_FAILURE_MODES.md). Live edits
committed: 2 of 6 to 5 of 6.

OPENCLAW LIVE

20.0 s edit, 33.4 s brief

real tool calls, v2 at 13:43 CDT; brief to v1 at
14:08 CDT; gated test 38.9 s, 7 tool calls
(RESULTS.md, BUILD_STATUS.md 17).

LATENCY, LOOPBACK

0.46 ms RTT, 0.78 ms tick

p50 idle, scripted controllers (p95 0.74 and
1.15 ms; under 2 x 30 inputs/s tick p95 12.3
ms; LATENCY.md 20:26Z). Not input to
photon.

PHONES

not yet with devices

controller page under mobile emulation and
23 integration tests with real sockets pass
(BUILD_STATUS.md 12); no physical phone
joined today.

OFFLINE PROOF

not run

only a before record exists
(offline-proof/1791049273707.json: egress
not blocked, server not up). Procedure:
docs/OFFLINE_PROOF.md.

Streaming, extension request to commit: 10.6 to 65.9 s, 4 of 6 committed, GPU contended (ACCEPTANCE.md, run 3 PASS 21 of 21: 56.0 s and 10.6 s); 9.5 s once on a quiet GPU (RESULTS.md). Trigger to request under 0.3 s; player, relic and score kept.

BEETLE 2D, PROMPT TO PLAYABLE 6 of 6 prompts to a validated, bot-playtested spec in 10.7 to 16.9 s (model 10.3 to 12.7 s), first model call, 0 repairs, 6 genres. Direct Ollama, shared GPU, Node harness: asset painting, worker hop and the browser not included (RESULTS.md 16:04 CDT).

Direct-harness timings are labelled [direct] in every report; OpenClaw timings are from the real server after the Ollama daemon restart at 13:43 CDT. Each card names its GPU state because the daemon was shared for most of the day. Mode briefs were measured at 14:25 to 14:42 CDT in direct mode with no controllers connected (docs/RESULTS.md). The 12 cards are the 3D engine; the 2D line is the 2D studio.

Business hypothesis and next validation



HYPOTHESIS

- Studios will pay for faster prototype iteration that keeps playtests running.
- Their data stays on their machines, which matters to studios with unreleased IP.

NOT CLAIMED

- Market size.
- Pricing.
- Customer traction.
- Speed-up percentages without a measured baseline.

NEXT VALIDATION

- Time a conventional edit-rebuild-regather loop with the same designers and the same change list.
- Compare it against Beetle's measured edit-to-commit times.



Change the game without stopping the game.